

Lab: Exploring Cognition

Objective

Design and simulate a simple agent that demonstrates the principles of **Cognition**.

Prerequisites

- Basic understanding of Python or a relevant simulation tool.
- Familiarity with the formal definition of Cognition.

Steps

1. **Define the Environment:** Create a simple grid world or state space relevant to Robotics.
2. **Define the Agent:** Specify the agent's generative model, focusing on Cognition.
3. **Simulation:** Run the agent for 100 timesteps.
4. **Perturbation:** Introduce a specific challenge to Cognition (e.g., increased noise, occlusion).
5. **Analysis:** Plot the Free Energy over time. Does the agent successfully adapt?

Discussion Requirements

- Attach your code or simulation logs.
- Explain the specific mechanism used to implement Cognition.